2 – Reconnaissance Practice Report

Reconnaissance is the first stage of the Vulnerability Assessment and Penetration Testing (VAPT) process. During this phase, publicly available information about the target system is collected without actively exploiting the system. The objective is to identify domain information, DNS records, subdomains, and open network services.
In this phase, reconnaissance activities were conducted using Kali Linux with various command-line tools to gather information about the target domain example.com.

Tools Used

The following tools were used during the reconnaissance phase:

- **WHOIS**
- **Dig**
- **Sublist3r**
- **Nmap**

These tools help security professionals gather information about target systems during the early stages of security assessment.

- **WHOIS Lookup:** To gather domain registration details such as registrar, creation date, and name servers.

Tool Used

- **WHOIS**

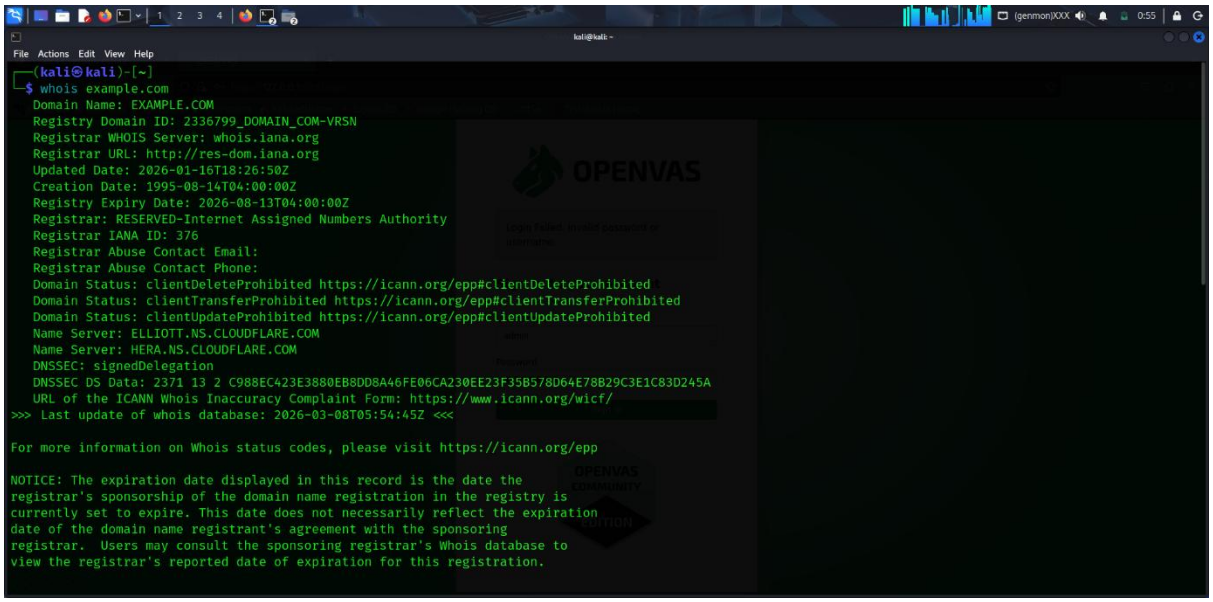
Command

- **whois example.com:** The WHOIS lookup retrieves domain registration information from global domain registries.

Key Findings

Field	Result
Domain	example.com
Registrar	IANA
Creation Date	1995-08-14
Expiry Date	2026-08-13
Name Servers	elliott.ns.cloudflare.com
	hera.ns.cloudflare.com

These results indicate that the domain uses Cloudflare DNS infrastructure.



```
(kali㉿kali)-[~]
└─$ whois example.com
   Domain Name: EXAMPLE.COM
   Registry Domain ID: 2336799_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.iana.org
   Registrar URL: http://res-dom.iana.org
   Updated Date: 2026-01-16T18:26:50Z
   Creation Date: 1995-08-14T04:00:00Z
   Registry Expiry Date: 2026-08-13T04:00:00Z
   Registrar: RESERVED-Internet Assigned Numbers Authority
   Registrar IANA ID: 376
   Registrar Abuse Contact Email:
   Registrar Abuse Contact Phone:
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Name Server: ELLIOTT.NS.CLOUDFLARE.COM
   Name Server: HERA.NS.CLOUDFLARE.COM
   DNSSEC: signedDelegation
   DNSSEC DS Data: 2371 13 2 C988EC423E3880EB8DD8A46FE06CA230EE23F35B578D64E78B29C3E1C83D245A
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-03-08T05:54:45Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar.  Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.
```

WHOIS lookup results for example.com

- **DNS Enumeration using DIG:** To identify the IP addresses associated with the target domain.

Tool Used

- **DIG**

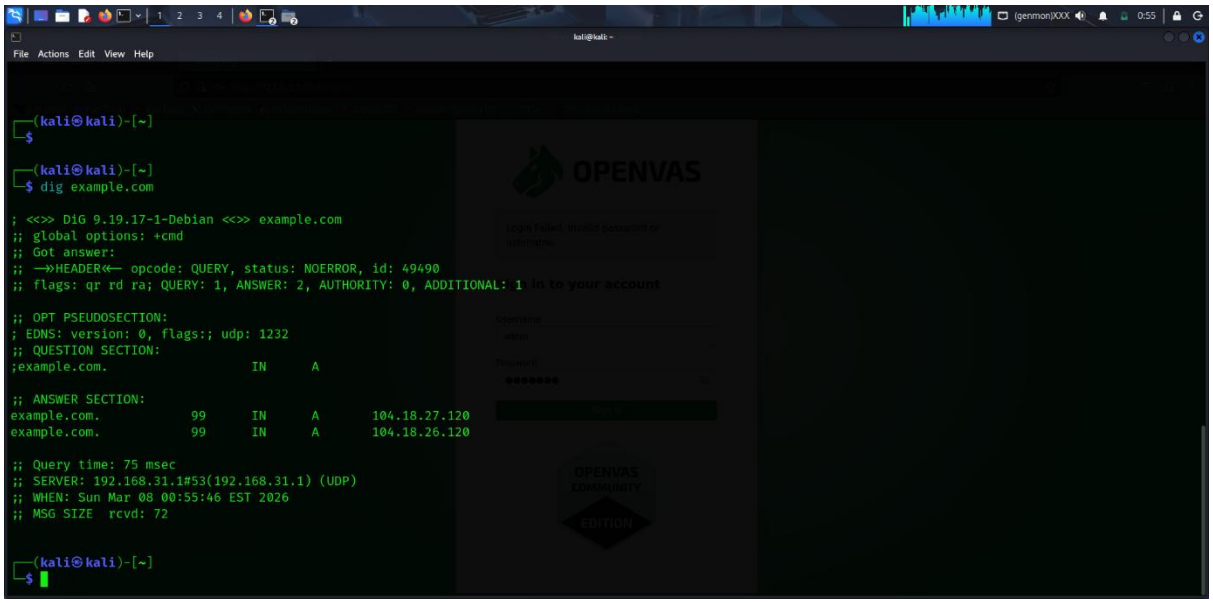
Command

dig example.com: The DIG command retrieves DNS records from domain name servers.

Key Findings

Record Type	IP Address
A Record	104.18.27.120
A Record	104.18.26.120

These IP addresses belong to Cloudflare's reverse proxy network, which protects the original web server.



DNS lookup results using DIG command

- **Subdomain Enumeration:** Identify additional subdomains associated with the target domain.

Tool Used

- **Sublist3r**

Command

- **sublist3r -d example.com**: Sublist3r performs passive reconnaissance by querying multiple public sources such as:

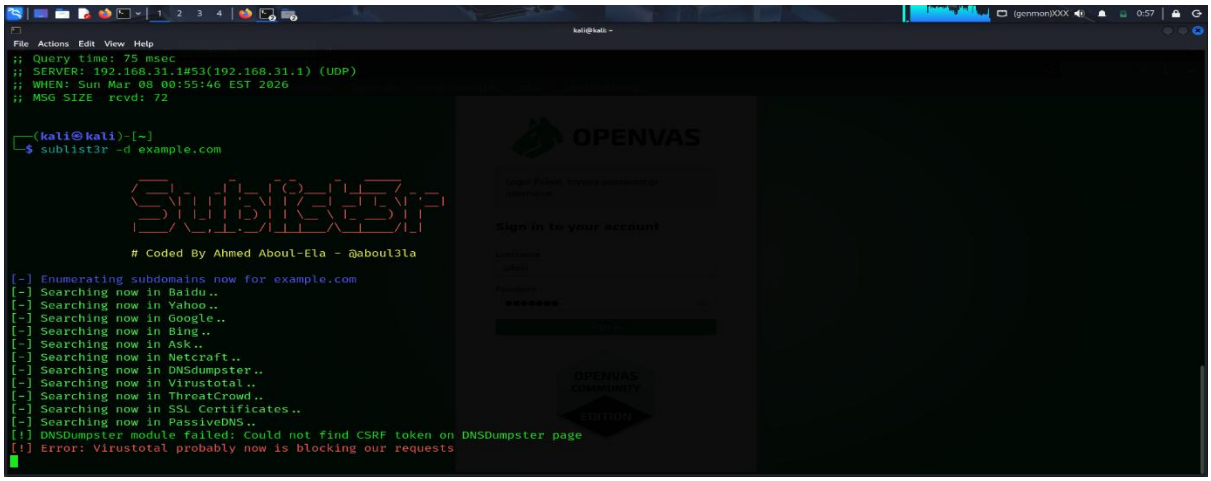
✓ Google
✓ Bing
✓ Yahoo
✓ Netcraft
✓ VirusTotal
✓ SSL Certificates
✓ PassiveDNS

During enumeration, the tool attempted to query multiple data sources.

Some modules returned warnings:

- DNSDumpster module failed
- VirusTotal blocked automated requests

These warnings are common because some platforms block automated scanning requests.



Subdomain enumeration using Sublist3r

- Network Scanning: Identify open ports and running services on the target system.

Tool Used

- Nmap

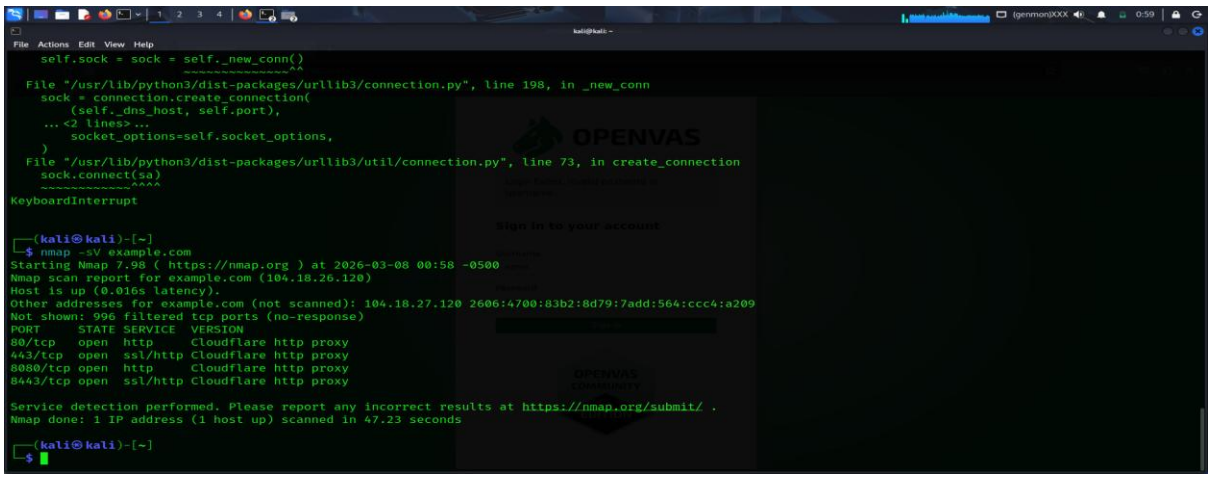
Command Used

- **nmap -sV example.com:** The Nmap scan identifies open ports and the services running on those ports.

Scan Results

Port	State	Service	Version
80	Open	HTTP	Cloudflare HTTP Proxy
443	Open	HTTPS	Cloudflare HTTP Proxy
8080	Open	HTTP	Cloudflare Proxy
8443	Open	HTTPS	Cloudflare Proxy

The scan results indicate that the target domain is protected by Cloudflare reverse proxy infrastructure, which masks the actual web server.



Nmap service detection scan results

➢ Vulnerability Tracking Table

During the reconnaissance phase, the discovered services were documented for further vulnerability analysis.

Scan ID	Vulnerability	CVSS Score	Priority	Host
001	Open HTTP Port	5.0	Medium	example.com
002	Open HTTPS Port	4.5	Medium	example.com

This table will be expanded during the vulnerability scanning phase.

Analysis

The reconnaissance activities successfully gathered important information about the target domain. The WHOIS lookup revealed domain registration details and DNS infrastructure. DNS enumeration identified the IP addresses associated with the domain, which belong to Cloudflare. Subdomain enumeration attempted to identify additional subdomains using passive reconnaissance techniques. Finally, the Nmap scan detected open ports and running services.

The results indicate that the target website uses Cloudflare protection, which hides the origin server and acts as a reverse proxy for web traffic.

Conclusion

The reconnaissance phase successfully collected publicly available information about the target domain. By using tools such as WHOIS, DIG, Sublist3r, and Nmap, it was possible to identify domain registration details, DNS records, subdomains, and open network services.

This information will be used in the next phases of the penetration testing process to perform vulnerability scanning and exploitation analysis.